Federico Mazzoni

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AI Engineer with experience delivering production LLM systems, RAG pipelines, agentic workflows and automation systems. Owned AI products from requirements through AWS deployment, improving customer engagement and reducing support workload and costs. Research experience in AI spanning adaptive retrieval, human-AI collaboration, and NLP, with 8 peer-reviewed papers. **EU citizen based in Ireland.**

Experience

AI Engineer Aug. 2026 – Present
Marriott International *Cork, Ireland*

- **Multi-agent Assistant:** Developed a multi-agent solution integrated with legacy in-house systems and Salesforce data, featuring RAG-based Q&A, human escalation, and advance booking capabilities.

AI & Automation Engineer May 2024 – July 2026
Domotz *Dublin, Ireland*

- **Chatbot:** Engineered a FastAPI RAG-based chatbot (Qdrant, LangChain, PostgreSQL) deployed on AWS, boosting customer engagement by 65% and resolving 25% of tickets. Built multiple knowledge bases and UIs.
- **Dynamic MCP:** Developed an MCP server acting as a semantic proxy across 10+ MCP servers, dynamically providing only the tools and skills relevant to each user's task via hybrid semantic search, reducing agent context.
- **Conversation Intelligence:** Built a Gong alternative for Sales & CS teams, with classification, summarization, and chat over thousands of emails, tickets, and call transcripts, saving ∼$50k/year in subscription costs.
- **Automation Systems:** Built automation systems that reduced sales, finance, and marketing workflows from days to hours, including a Slack sales bot and invoice processors handling hundreds of customers.
- **Anomaly Detection:** Combined traditional ML and LLM reasoning to detect network anomalies in time series.

PhD Research Fellow in AI Nov. 2022 – May 2024
University of Pisa / National Research Council of Italy *Pisa, Italy*

- **Research Focus:** Collaborative human-AI systems, algorithmic fairness, linguistic trust in social networks.
- Published peer-reviewed articles, managed multiple concurrent projects from concept to validation, mentored university students, presented research at conferences, and organised the *Law, Society, and AI* workshops.

Freelance Data Analyst Jan. 2022 – Nov. 2022
Self-Employed *Pisa, Italy*

- Built recommendation systems, scraped web data for datasets, and explored bioinformatics biomarker methods.

Personal Projects

iRAG: AI system that learns from human decisions, reducing error while gradually taking on more autonomy.
GAVEL: Ethical evaluation of AI model behaviour and decision-making using Game Theory.
StockAgent: Multi-agent AutoGen system for stock market analysis with a custom MCP yfinance server.
HINTT: Real-time OCR+translation tool sending screenshots to OpenAI/Ollama. Employed tkinter and mss.
Simple Palette Maker: Streamlit app to colorize black-and-white GameBoy palettes. 1k+ users.

Skills

AI/ML: LLMs (LangChain, LangGraph, AutoGen), RAG, HuggingFace, Scikit-Learn, PyTorch, RiverML
Backend & Deployment: FastAPI, Streamlit, Docker, AWS (EC2, S3, CloudWatch, Terraform), PostgreSQL, Qdrant
Platforms: Git, Jira, BitBucket, Snowflake, Zendesk, Salesforce, n8n, OpenAI, Claude, Amazon Bedrock, Azure
Programming: Python, JavaScript, R · **Languages:** English (C1), Italian (Native), German (A1)

Education

University of Pisa 2020 – 2022
Master's Degree in Digital Humanities (Data Science and NLP) — 110/110 with honours (GPA: 4/4) *Pisa, Italy*

- Relevant Courses: Data Mining, Social Network Analysis, Text Analytics, Distributed Data Analysis and Mining, Semantic Web, Laboratory of Data Science, Statistics, Applied Linguistics, Computational Linguistics.

University of Pisa 2014 – 2017
Bachelor's Degree in Philosophy — 110/110 with honours (GPA: 4/4) *Pisa, Italy*